

Item 3b: Model Comparison & Evaluation

Below is the evaluation of the 4 models when I asked the question *"Is this course hard?"* using the course syllabus document:

1. Output Summaries & Results

- **gpt-3.5-turbo:**
 - **Behavior:** Returned a very high-level, generic paragraph about Python and AI concepts. It lacked specific syllabus details (like grading distribution, lab deadlines, or project requirements).
- **gpt-4.1:**
 - **Behavior:** Returned a highly structured, accurate, and comprehensive response. It cited specific policies directly from the syllabus (e.g., Thursday lab deadlines, Sunday homework deadlines, 25% exam, 25% group project, \$10–\$30 API cost estimate) and provided a balanced verdict based on prior Python knowledge.
- **gpt-5-chat-latest (gpt-5-mini):**
 - **Behavior:** gpt-5-chat-latest returned an API error (404 – model_not_found), as it is not available on the current API access; gpt-5-mini was therefore used in its place in the deployed app. gpt-5-mini returned a fast, well-structured response with clear sections ("Why it might feel hard," "Why it might feel easy," "Estimated time commitment," "Practical tips to succeed"). It accurately captured key syllabus specifics — no-late-submission labs, the 25%-per-day homework penalty, the closed-book 25% exam, and the team project involving RAG and tooling — and estimated the time commitment (~3 hrs in-class, 4–8 hrs/week outside).
- **gpt-5-nano:**
 - **Behavior:** Returned a fast, well-structured response with clear bullet points ("Why it's challenging", "Who might find it harder", "Tips to succeed"). It captured key specifics (9 labs, 7 homeworks, team project details) accurately.

Which Answer is Best? (Quality, Cost & Speed Evaluation)

- **Best Overall Quality Answer: gpt-4.1**

- **Reasoning:** gpt-4.1 provided the most precise synthesis of the syllabus. It correctly identified every nuance—from grading weights and late policies to specific technical concepts (Streamlit, RAG, APIs)—and gave a clear, actionable conclusion for the student.
- **Best Answer Considering Cost, Speed, and Efficiency: gpt-5-nano**
 - **Reasoning:** gpt-5-nano is significantly faster and lower cost per query than gpt-4.1. Despite its smaller footprint, it extracted all critical syllabus facts (lab counts, homework counts, project percentages, API costs) and formatted them concisely without missing key details.
- **Conclusion:**

gpt-4.1 is the best model for overall quality and depth of analysis. However, when factoring in **cost and speed**, **gpt-5-nano** is the best choice for everyday Q&A applications because it delivers nearly identical accuracy at a fraction of the response time and latency cost. (gpt-3.5-turbo lacked sufficient detail, and gpt-5-chat-latest was unavailable on the current API access.)

References:

1. **Gemini:** Used understand to understand code and make changes according to get the desired output.